

CSE370 : Database Systems
Project Report
Project Title : EcoTrade:Buy and Sell Used Goods

Group No : _16_, CSE370 Lab Section : _01_, Fall 2024		
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Introduction

Eco Trade is a dynamic web application built using PHP, HTML, CSS, and MySQL, designed to support the eco-friendly buying and selling of second-hand products. This project encourages sustainability by promoting the reuse of items rather than discarding them. It serves as a user-friendly platform where buyers and sellers can connect securely and efficiently. Developed as part of a university-level database management systems (DBMS) course[CSE370], the project also focuses on backend logic and practical implementation of database operations.

Project Features

User Registration and Login
Users can create accounts and securely log in using PHP session handling.

Combined Login and Registration Form
A single page handles both login and sign-up processes with fields for phone number, address, zip code, etc.

Homepage Product Display
The homepage lists all products from the database with filtering options and a clean UI inspired by bikroy.com.

Become a Seller
Registered users can switch to seller mode and post their products with details including name, price, description, category, phone number, and image.

Add to Cart Functionality (PHP only)
Buyers can add products to their cart using PHP (no JavaScript). Cart items are stored and retrieved from the MySQL database.

Cart Page
Buyers can view, update, or remove items from their cart. Total price and quantity are shown.

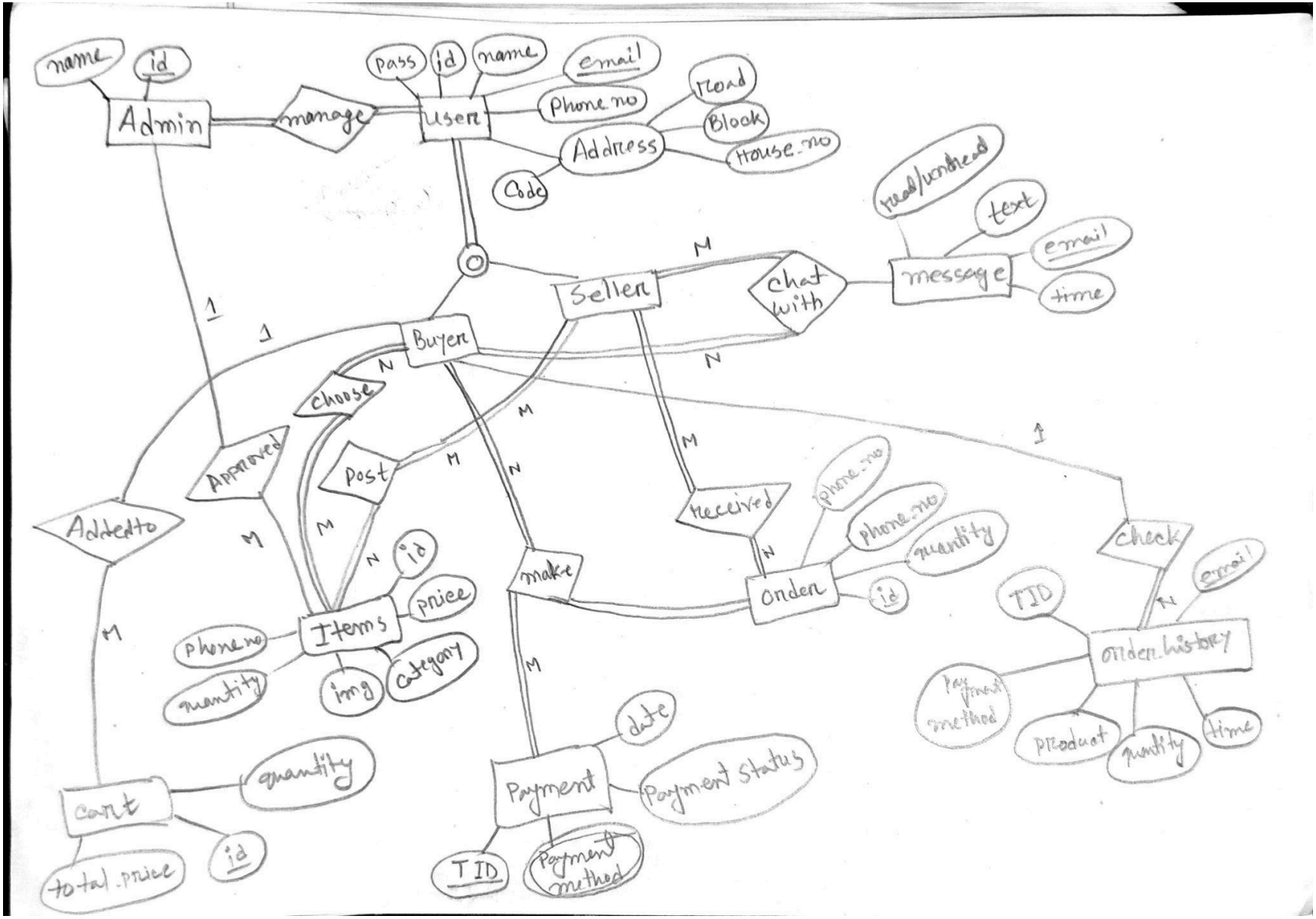
Order Placement Page
A structured form collects detailed shipping and contact information including house no, road no, block, city, district, division, and recipient name.

Payment Page
After order submission, users are taken to a payment page where they choose the payment method (Cash on Delivery, Visa, or Bkash) and provide a transaction ID. Payment status is stored.

Order History Page
Users can track their previous purchases. Information includes product name, quantity, transaction ID, payment method, and timestamp.

Chat System
A simple messaging feature allows buyers and sellers to communicate directly within the platform.

ER/EER Diagram


name
id
Admin
manage
pass
id
name
email
User
Phone no
road
Block
House_no
Address
Code
read/unread
text
email
time
message
chat with
Seller
Buyer
choose
Approved
Post
Add to
Items
id
price
Phone no
quantity
img
category
make
received
Phone_no
Phone_no
quantity
Order
id
check
email
TID
Order_history
Payment method
Product
quantity
time
Cart
quantity
id
total_price
Payment
date
Payment Status
TID
Payment method

Schema Diagram


Admin
name
id
Users
pass.
id
fname
lname
email
phone_no
Zip
road
Block
HouseNo
id
Product
id
user_email
product_name
price
phone
description
category
image
Cart
id
user_email
product_id
quantity
Payment_method
Paymentmethod
T.ID
Payment
id
user_email
phone
T.ID
payment_status
time
Order_history
id
user_email
productname
quantity
total price
T.ID
Chat_messages
id
sender_email
receiver_email
message
time
status

Frontend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 22299383, Name : Pallab Saha

Eco Trade begins at **register.php**, where new users can **sign up** by providing essential information such as name, email, password, phone number, and address. Once registered, users are redirected to the login page, where they can securely log into their accounts. Logging in takes them to the main interface, **homepage.php**, which features a clean layout and a personalized welcome message displaying the user's name.

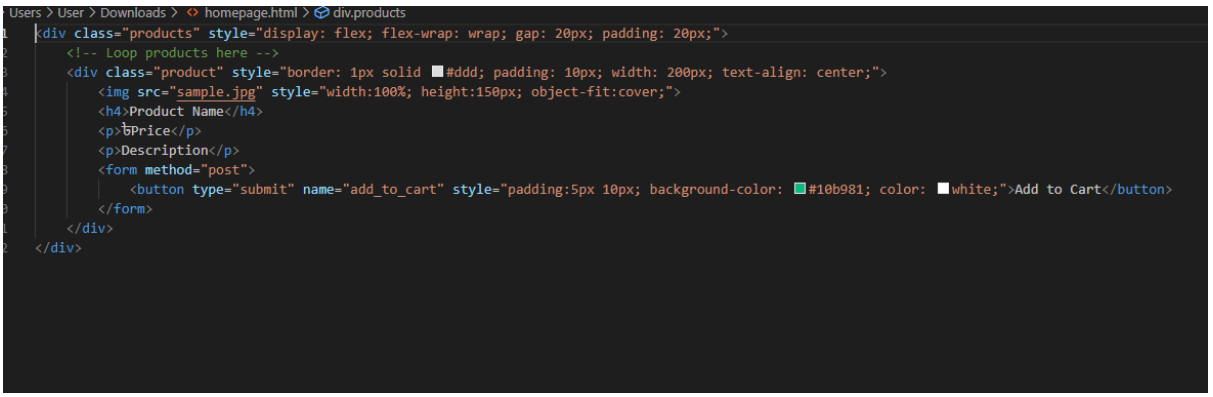
At the top of every page, a consistent navigation bar (navbar) is displayed, which includes the Eco Trade logo, a search bar, and quick access links to Seller, Cart, Orders, Profile, Chat, and Logout (via **logout.php**). Dynamic badges indicate the number of items in the cart and new messages in the chat system.

The homepage.php also includes **category filter buttons** (like Electronics, Books, Vehicles, etc.) and a responsive grid of products. Each product card displays an image, name, price, category, and a phone number, along with an "**Add to Cart" button** that connects with **cart.php.**

Sellers can post their items through **products.php,** which provides a user-friendly product submission form. This form allows input of the product name, price, contact phone number, category, and image. It's styled using CSS for better responsiveness and clarity.

In **cart.php**, users can view all added items in a tabular layout, showing the product name, quantity, price, and providing options to remove products or proceed to payment.

```
<h2>Your Cart</h2>
<table border="1" style="width:80%; margin:auto;">
    <tr><th>Product</th><th>Quantity</th><th>Price</th><th>Action</th></tr>
    <tr>
        <td>Product A</td><td>1</td><td>৳100</td>
        <td><button>Remove</button></td>
    </tr>
</table>
<form action="order.php" method="post" style="text-align:center; margin-top:20px;">
    <button type="submit" style="padding:10px 20px; background-color: green; color: white;">Order Now</button>
</form>
```

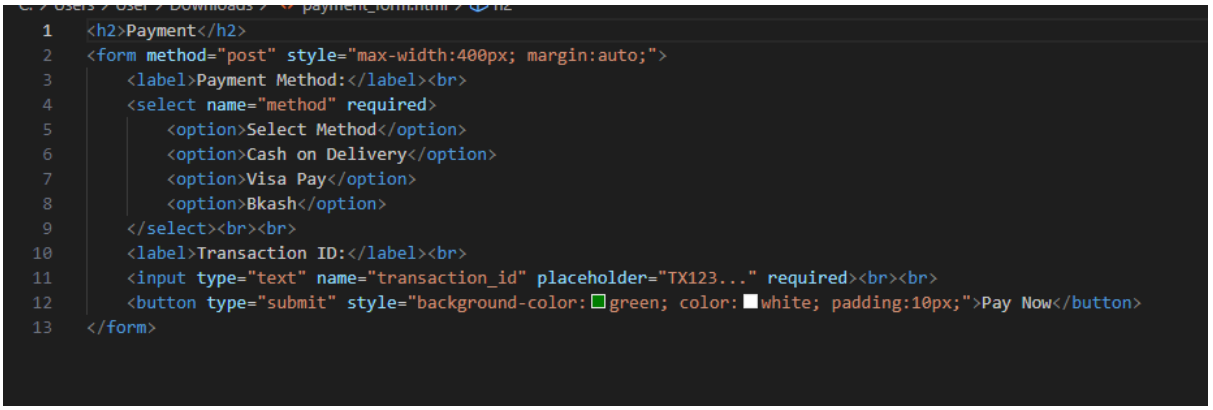


Contribution of ID : 22299227, Name : Avrow Jothi Bose

The **payment.php** page allows users to finalize their purchase by selecting a payment method such as Cash on Delivery, Visa Pay, or Bkash. A transaction ID can be entered, and the payment status is shown after submission.

After completing a purchase, users can view all their past transactions in **order_history.php**. This page lists product names, quantities, payment method, transaction ID, and the time of purchase in an organized table layout.

Lastly, users can access **chat.php** to communicate in real-time. The navbar shows a red badge if there are unread messages, enhancing user interaction and platform engagement.





Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 22299383, Name : Pallab Saha

1.User Profile (`profile.php`)

- Gets user data from the users table using session email.
- Shows:
    - User's name, email, phone number, address, etc.
- Backend ensures:
    - Only logged-in users can see their profile.
    - Session is used to securely fetch personal info.

2.Product Description (`product_description.php`)

- Gets product info from database using product ID (via URL).
- Shows:
    - Image, price, description, category
    - Seller’s full name from users table

- Lets logged-in users:
    - Add product to cart from this page.

3.Homepage (`homepage.php`)

- Fetches all available products from the `products` table.
- Displays:
    - Product name, price, image, short description
- Logged-in users can:
    - View product details
    - Add items to cart directly
- Backend also uses:
    - Session to keep track of logged-in user
    - `$_POST` to handle add-to-cart requests

4.Cart System (`cart.php`)

- Shows all items added to the user's cart.
- Displays:
    - Product image, name, price, quantity
- Calculates and shows Grand Total.
- Lets users remove items from cart.

- Sends data to payment page.

Contribution of ID : 22299227, Name : Avrow Jothi Bose

In our project, we used PHP and MySQL for backend development.

Main Roles of Backend in Our Project:

1.Database Connection (`connect.php`)

- Connects the website to the database.
- It helps us store and get data like user info, products, and orders.

2. Payment Handling (`payment.php`)

- When a user confirms a payment:
    - A unique transaction ID is generated.
    - Payment details are saved in the database.
    - Products from the cart are saved in the order history.
    - The cart is cleared after successful payment.

3.Order History (`order_history.php`)

- Shows the user all their previous orders.
- It displays product names, quantity, payment method, and transaction ID.
- Data is fetched from the `order_history` and `payment` tables.

4.Session Handling

- After login, the user's email is saved in a session (`$_SESSION`).

- If the user is not logged in, they are redirected to the login page.
- This keeps the system secure and private.

5. Personal Chat System (chat.php)

- Allows users to **send and receive messages** in real time.
- Stores each message in the **`messages`** table with sender, receiver, message text, and timestamp.
- Helps buyers and sellers communicate about products.
- Makes the website more interactive and user-friendly.

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

CSE370 Project final

Conclusion

Eco Trade successfully integrates core PHP and MySQL functionalities to deliver a sustainable e-commerce solution. It provides all essential features of a modern buy-and-sell platform while promoting environmental awareness through product reuse. This project not only fulfills academic objectives but also reflects real-world application of backend development and database management. With future improvements like live notifications, user reviews, and enhanced security, the platform can become even more robust and scalable.

References

PHP Full Course for non-haters 🐘

Login & Register Web Form using PHP XAMPP Frontend, Backend & D…

https://youtube.com/playlist?list=PLzzYtJMBTYCeQLDAUJKIhHN8TB8KGsDES&si=MfX_fvNL5vRuOZoe

https://youtube.com/playlist?list=PLEcb0EsInoSmx15xhjWjqcSZTg86GUvoo&si=ySJjZ9vsviBhfbn3